

THE COPRIME BURATTI–HORAK–ROSA CONJECTURE HOLDS FOR THE SUPPORT $\{1, 6, 18\}$

ABSTRACT. Ağirseven and Ollis proved the Coprime Buratti–Horak–Rosa Conjecture for the support $\{1, 6, 18\}$ except possibly at $v \in \{47, 49, 59\}$. We close these three orders by exact finite certificates: a deterministic search, specified in full in Section 2, returns one Hamiltonian path for every one of the 4,123 nonnegative multiplicity triples at these orders, including all 3,667 triples with support exactly $\{1, 6, 18\}$, and every path it returns was checked in exact integer arithmetic. Consequently, the Coprime Buratti–Horak–Rosa Conjecture holds for this support.

1. RESULT

Label the vertices of K_v by $\{0, 1, \dots, v-1\}$ and define the cyclic length of an edge by

$$\ell_v(x, y) = \min\{|x - y|, v - |x - y|\}.$$

A Hamiltonian path $P = (p_0, p_1, \dots, p_{v-1})$ realizes the multiset

$$\Delta_v(P) = \{\ell_v(p_i, p_{i+1}) : 0 \leq i \leq v-2\},$$

with multiplicities retained. The arbitrary-order conjecture was formulated by Horak and Rosa [2]. Its coprime specialization, introduced by Ağirseven and Ollis [1, Conjecture 1.3], asserts that a multiset L of size $v-1$ is realizable whenever

$$\text{supp}(L) \subseteq \left\{1, 2, \dots, \left\lfloor \frac{v}{2} \right\rfloor\right\} \quad \text{and} \quad \gcd(v, x) = 1 \quad \text{for every } x \in L.$$

Theorem 1. *Let $a, b, c \geq 1$, put $v = a + b + c + 1$, and suppose that*

$$18 \leq \left\lfloor \frac{v}{2} \right\rfloor \quad \text{and} \quad \gcd(v, 18) = 1.$$

Then there is a Hamiltonian path P in K_v such that

$$\Delta_v(P) = \{1^a, 6^b, 18^c\}.$$

Equivalently, the Coprime BHR Conjecture holds for the support $\{1, 6, 18\}$.

Proposition 2. *Let $v \in \{47, 49, 59\}$ and let $a, b, c \in \mathbb{Z}_{\geq 0}$ satisfy $a + b + c = v - 1$. Then there is a Hamiltonian path P in K_v such that*

$$\Delta_v(P) = \{1^a, 6^b, 18^c\}.$$

An exponent equal to zero means that the corresponding length is omitted.

2020 *Mathematics Subject Classification.* 05C38, 05C78.

Key words and phrases. Buratti–Horak–Rosa conjecture, Hamiltonian path, cyclic edge length, computer-assisted proof.

Proof of Theorem 1. Ağirseven and Ollis prove the assertion for this support except possibly at $v = 47, 49, 59$ [1, Theorem 5.3]. Proposition 2 settles those three orders. $\square$

What is new here is exactly those three orders; the remaining orders for the support $\{1, 6, 18\}$ are due to Ağirseven and Ollis [1, Theorem 5.3] and are not reproved.

2. CERTIFICATE VERIFICATION

For $v \in \{47, 49, 59\}$, set

$$T_v = \{(a, b, c) \in \mathbb{Z}_{\geq 0}^3 : a + b + c = v - 1\}, \quad T_v^+ = T_v \cap \mathbb{Z}_{\geq 1}^3.$$

Thus $|T_v| = \binom{v+1}{2}$ and $|T_v^+| = \binom{v-2}{2}$. The certificate census is

v	$ T_v $	$ T_v^+ $
47	1128	990
49	1225	1081
59	1770	1596
Total	4123	3667

Proof of Proposition 2. Write $\hat{g} = \min\{g, v - g\}$. Two reductions organize the search. First, if $\gcd(v, s) = 1$ then multiplication by s is an automorphism of $\mathbb{Z}_v$, so a Hamiltonian path P realizes L if and only if sP realizes $\{\widehat{s\hat{x}} : x \in L\}$. Second, if a Hamiltonian path satisfies $|p_i - p_{i+1}| \leq \lfloor v/2 \rfloor$ for $0 \leq i \leq v - 2$, then $\Delta_v(P)$ is the multiset of those $v - 1$ differences; call such a path a *linear* realization. For $s = 6^{-1}$, $s = 18^{-1}$ and $s = 1$ the multiset $\{1^a, 6^b, 18^c\}$ becomes

	$s = 6^{-1}$	$s = 18^{-1}$	$s = 1$
$v = 47 :$	$\{8^a, 1^b, 3^c\}$	$\{13^a, 16^b, 1^c\}$	$\{1^a, 6^b, 18^c\}$
$v = 49 :$	$\{8^a, 1^b, 3^c\}$	$\{19^a, 16^b, 1^c\}$	$\{1^a, 6^b, 18^c\}$
$v = 59 :$	$\{10^a, 1^b, 3^c\}$	$\{23^a, 20^b, 1^c\}$	$\{1^a, 6^b, 18^c\}$

respectively, and every length displayed is at most $\lfloor v/2 \rfloor$.

For each v and each $(a, b, c) \in T_v$ we search for a linear realization of the first multiset of the row, then of the second, then of the third. The search starts the path at t , for $t = 0, 1, \dots, v - 1$ in turn, and extends it from its last vertex p to $p \pm d$, where d is a length whose multiplicity is not yet used up and $p \pm d \in \{0, 1, \dots, v - 1\}$ is not already on the path; candidates are tried in increasing order of the number of extensions they themselves admit, ties in increasing order of label, and the search backtracks when a vertex admits none. A partial path is abandoned when the vertices off it, together with p , fail to induce a connected subgraph of the graph that joins two elements of $\{0, 1, \dots, v - 1\}$ whenever they differ by a length still available, or when two vertices other than p have at most one neighbour in that subgraph; a partial

path that extends to a linear realization has neither defect. Each starting vertex is allotted 10^5 nodes of the search tree.

The search returns a linear realization for every one of the 4,123 targets: 3,792 are settled by the first multiset of their row, 43 by the second and 288 by the third. The largest single target took 1.51×10^7 nodes and the whole computation 4.11×10^9 . Multiplying by s^{-1} modulo v turns each linear realization into a Hamiltonian path in K_v , and each of the 4,123 paths was then checked in exact integer arithmetic to be a permutation of $0, 1, \dots, v-1$ whose $v-1$ cyclic lengths are exactly a ones, b sixes and c eighTEENS. $\square$

At the corners of T_v the certificates are closed forms: $p_i = i$, $p_i = 6i$ and $p_i = 18i$, reduced modulo v , realize $\{1^{v-1}\}$, $\{6^{v-1}\}$ and $\{18^{v-1}\}$ respectively. For $v = 59$ and $(a, b, c) = (13, 6, 39)$, one of the ten triples Ağırseven and Ollis list as not ruled out at $v = 59$ [1, Section 5], the search returns

0, 1, 2, 3, 4, 5, 6, 9, 12, 15, 18, 8, 11, 14, 17, 7, 10, 13, 16, 19,
22, 25, 28, 31, 21, 24, 27, 30, 20, 23, 26, 29, 32, 35, 38, 41, 51, 48, 58, 55,
45, 42, 52, 49, 46, 56, 53, 50, 47, 57, 54, 44, 34, 37, 40, 43, 33, 36, 39,

a linear realization $x_0, x_1, \dots, x_{58}$ of $\{1^6, 3^{39}, 10^{13}\}$, and $p_i = 6x_i \bmod 59$ then realizes $\{1^{13}, 6^6, 18^{39}\}$. The 4,123 paths are not printed one by one; the search specified above is the construction that returns them.

REFERENCES

- [1] O. Ağırseven and M. A. Ollis, *A Coprime Buratti–Horak–Rosa Conjecture and Grid-Based Linear Realizations*, arXiv:2412.05750v2 [math.CO], 2025. arXiv:2412.05750v2.
- [2] P. Horak and A. Rosa, *On a problem of Marco Buratti*, Electron. J. Combin. **16** (2009), no. 1, #R20. doi:10.37236/109.